Hunting for Habitable Worlds: A Transit Analysis of the STAC-1, STAC-2 and STAC-3 Systems

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STAC WARS, Problem Statement 3: Exoplanet Hunters

***Abstract*—We studied the brightness records of three stars, STAC-1, STAC-2 and STAC-3, looking for the small, repeating dips that a transiting planet produces. All three stars show them. From the spacing of the dips we find orbital periods of 4.2, 80 and 40 days, and from their depth we estimate planet radii of 5.98, 2.76 and 1.59 Earth radii. Kepler's third law places the planets at 0.051, 0.337 and 0.182 AU. We also estimated the transit probability, transit duration, stellar flux and equilibrium temperature of each planet. None of the planets lies inside its star's habitable zone, but STAC-3 b, a super-Earth orbiting a cool M-type star, comes closest and is the planet we would choose for follow-up.**

***Index Terms*—exoplanets, transit photometry, Kepler's third law, habitable zone, equilibrium temperature**

I. Introduction

Most of the exoplanets known today have never been seen directly. They were found because the star they orbit became very slightly fainter, for a few hours, every time the planet passed in front of it. This is the transit method, and the Kepler and TESS missions have used it to find thousands of planets [1], [2].

In this problem we were given brightness measurements of three stars and asked to (A) decide whether each one hosts a planet, (B) estimate the planet's size, (C) work out its orbit and (D) pick the most promising world for further study.

II. Theory

A. Transit Depth

If a planet of radius R_p crosses the disc of a star of radius R_s, it covers an area πR_p^2 out of a total πR_s^2. If the stellar disc is equally bright everywhere, the fraction of light lost is

$$\delta = (F_0 - F_t) / F_0 \approx (R_p / R_s)^2 \tag{1}$$

where F_0 is the normal flux and F_t the flux during transit [3]. Rearranging gives $R_p = R_s\sqrt{\delta}$. The problem sheet writes the ratio the other way round, $(R_s/R_p)^2$, but that would make δ larger than one, so we use the form above.

B. Kepler's Third Law

For a planet much lighter than its star on a circular orbit, gravity provides the centripetal force:

$$GM_s m / a^2 = m(2\pi/P)^2 a \;\Rightarrow\; P^2 = 4\pi^2 a^3 / (GM_s) \tag{2}$$

With P in years, a in AU and M_s in solar masses, the constants cancel and the law becomes

$$P^2 = a^3 / M_s \;\Rightarrow\; a = (P^2 M_s)^{1/3} \tag{3}$$

C. Transit Probability and Duration

We only see a transit if the orbit is lined up with our line of sight. For a randomly tilted circular orbit the probability is roughly

$$p_{\mathrm{tr}} \approx R_s / a \tag{4}$$

For a central transit the planet travels about $2R_s$ at a speed of $2\pi a/P$, so the transit lasts about [4]

$$T_{\mathrm{dur}} \approx (P / \pi)(R_s / a) \tag{5}$$

D. Habitable Zone and Temperature

Starlight spreads out as $1/d^2$. A planet receives the same energy as Earth when $L/d^2 = L_\odot/(1\ \mathrm{AU})^2$, which gives the simple habitable-zone distance

$$d_{\mathrm{HZ}} \approx \sqrt{(L / L_\odot)}\ \ \mathrm{AU} \tag{6}$$

More generally, the flux a planet receives relative to Earth, and its equilibrium temperature, are [5]

$$S = (L/L_\odot) / a^2, \quad T_{\mathrm{eq}} = 278.3\ \mathrm{K}\ (1 - A)^{1/4}\, S^{1/4} \tag{7}$$

where a is in AU and A is the Bond albedo. For Earth ($A = 0.3$) this gives about 255 K, before the greenhouse effect warms the surface.

III. Data and Assumptions

Table I lists the stellar properties we were given. STAC-1 was measured once a day for 21 days. STAC-2 was measured every 20 days and STAC-3 every 10 days, both over 200 days. Outside the dips the flux is always 1.0000, so we take F_0 = 1.0000.

TABLE I
Stellar Properties

	STAC-1	STAC-2	STAC-3
Mass ($M_\odot$)	1.00	0.80	0.50
Radius ($R_\odot$)	1.00	0.80	0.50
Radius (km)	696 000	556 800	348 000
Luminosity ($L_\odot$)	1.00	0.40	0.08
Temperature (K)	5800	5000	3800
Approx. type	G (Sun-like)	K	M (red dwarf)

We assumed circular orbits and central transits, that each low reading falls in the time slot that contains the middle of a transit, and that the small differences between dip depths (around 10^{-4}) are noise. We used $R_\odot$ = 696 000 km, $R_\oplus$ = 6371 km, 1 AU = 1.496×10^8 km and 1 yr = 365.25 d. All values were worked out by hand and then checked with a short Python script. Before trusting it on the STAC data, we tested the script on our own Solar System: it returns 1.000 AU for Earth, 0.387 AU for

Mercury and 5.20 AU for Jupiter, and 1.00 Earth radius for an Earth transiting the Sun.

IV. Analysis and Results

A. Part A: Detecting the Planets

STAC-1. The flux drops to 0.9970 on days 3, 8, 12, 16 and 20. Our first guess was a 4-day period, but the first gap is 5 days, so no whole number of days fits. Fitting a straight line $t_n = t_0 + nP$ to the five dip times (n = 0 to 4, so $\bar{n}$ = 2 and $\bar{t}$ = 11.8 d) gives

$P = \Sigma(n - \bar{n})(t_n - \bar{t}) / \Sigma(n - \bar{n})^2 = 42 / 10 = 4.2$ d

$t_0 = \bar{t} - P\bar{n} = 11.8 - 4.2 \times 2 = 3.4$ d

This predicts transits at 3.4, 7.6, 11.8, 16.0 and 20.2 d, which round to exactly the days on which we see dips. The depth is δ = (1.0000 − 0.9970)/1.0000 = 0.0030.

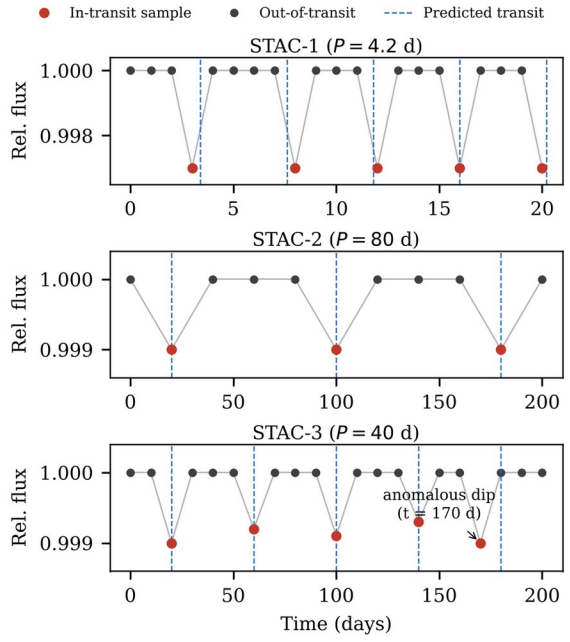


Fig. 1. Light curves of the three stars. Red points are in-transit readings; dashed lines are the transit times predicted by our periods. Note the unexplained dip of STAC-3 at 170 d.

STAC-2. Dips appear at 20, 100 and 180 d, each down to 0.9990. Both gaps are 80 d, so P = 80 d and δ = 0.0010. Because the star was only measured every 20 days, a period of 80/3 ≈ 26.7 d could also fit. A period of 40 d cannot, since it would need dips at 60 and 140 d. We keep 80 d as the simplest answer.

STAC-3. Dips at 20, 60, 100 and 140 d are exactly 40 d apart, giving P = 40 d. Their depths are 0.0010, 0.0008, 0.0009 and 0.0007, so the mean depth is δ = 0.0034/4 = 0.00085. There is one more dip at 170 d, but the 40-day pattern puts the next transit at 180 d, where the flux is normal. This could be a flare or starspot on this cool, active star, a second planet, or an instrument glitch. A single period of 38 d would also place all five dips inside the 10-day sampling slots, so the period is not completely certain. We keep 40 d because four dips line up exactly.

Why the dips point to a planet. A planet blocks the same fraction of the star once per orbit, so its dips repeat at a fixed interval and have the same depth each time. Flares make a star brighter, and starspots cause slow, smooth changes rather than short dips. Repeated, equal-depth dips are therefore the signature of a planet. Fig. 1 shows the light curves with our predicted transit times.

B. Part B: Planet Size

Using $R_p = R_s\sqrt{\delta}$ and dividing by $R_\oplus$:

STAC-1 b: 696 000 × √0.0030 = 696 000 × 0.05477 = 38 121 km = **5.98 $R_\oplus$**

STAC-2 b: 556 800 × √0.0010 = 556 800 × 0.03162 = 17 608 km = **2.76 $R_\oplus$**

STAC-3 b: 348 000 × √0.00085 = 348 000 × 0.02915 = 10 146 km = **1.59 $R_\oplus$**

With the Kepler size classes [1] (Earth-size below 1.25 $R_\oplus$, super-Earth 1.25–2, Neptune-size 2–6, Jupiter-size above 6), STAC-1 b is a large Neptune-size planet, STAC-2 b a small Neptune-size planet and STAC-3 b a super-Earth. Using the shallowest and deepest single dips of STAC-3 instead of the mean gives a radius between 1.45 and 1.73 $R_\oplus$, so it is a super-Earth either way. STAC-2 and STAC-3 have almost the same depth, yet STAC-3 b is much smaller, simply because its star is smaller.

C. Part C: Orbits

Converting each period to years and applying (3):

STAC-1 b: P = 4.2/365.25 = 0.01150 yr, a^3 = 0.01150² × 1.0 = 1.322×10⁻⁴, a = **0.0509 AU** = 7.62×10⁶ km

STAC-2 b: P = 0.2190 yr, a^3 = 0.04797 × 0.8 = 0.03838, a = **0.337 AU** = 5.05×10⁷ km

STAC-3 b: P = 0.1095 yr, a^3 = 0.01199 × 0.5 = 0.005997, a = **0.182 AU** = 2.72×10⁷ km

As a check, we recomputed STAC-3 b in SI units using (2). With GM_s = 6.674×10⁻¹¹ × 9.945×10²⁹ = 6.637×10¹⁹ m³ s⁻² and P = 40 × 86 400 = 3.456×10⁶ s, we get $a = (GM_sP^2/4\pi^2)^{1/3}$ = 2.72×10¹⁰ m, the same result.

All three planets are much closer to their stars than Earth is to the Sun. STAC-1 b is about 20 times closer, and even STAC-2 b is only at roughly Mercury's distance (0.39 AU). Close-in planets transit more often, so they are the easiest to find.

TABLE II
Summary of Results

	STAC-1 b	STAC-2 b	STAC-3 b
Period P (d)	4.2	80	40
Depth δ	0.0030	0.0010	0.00085
R_p (km)	38 121	17 608	10 146
R_p ($R_\oplus$)	5.98	2.76	1.59
Size class	Neptune-size	Neptune-size	Super-Earth
a (AU)	0.0509	0.337	0.182
p_{tr}	0.091	0.011	0.013
T_{dur} (h)	2.9	6.7	3.9
d_{HZ} (AU)	1.00	0.632	0.283
a / d_{HZ}	0.05	0.53	**0.64**
Flux S ($S_\oplus$)	385	3.5	**2.4**
T_{eq}, A = 0.3 (K)	1128	349	**318**

D. Transit Probability and Duration

From (4) and (5), STAC-1 b has p_{tr} = 696 000/7.62×10^6 = 0.091 and T_{dur} = (4.2/π) × 0.091 = 0.122 d ≈ 2.9 h. STAC-2 b gives p_{tr} = 0.011 and T_{dur} ≈ 6.7 h, and STAC-3 b gives p_{tr} = 0.013 and T_{dur} ≈ 3.9 h. So STAC-2 b and STAC-3 b only had about a 1 in 80 to 1 in 90 chance of being lined up with us at all, and every transit lasts just a few hours. With readings 10 to 20 days apart, a transit this short would usually fall between readings, so denser sampling is needed to confirm these periods. Table II collects all of our results.

V. Part D: The Most Promising World

Using (6), the habitable zone lies near 1.00 AU for STAC-1, 0.632 AU for STAC-2 and 0.283 AU for STAC-3. All three planets orbit inside these distances, so all three are hotter than Earth. The flux from (7) makes the difference clear. STAC-1 b receives 1.00/0.0509^2 ≈ 385 times Earth's flux (T_{eq} ≈ 1128 K), STAC-2 b receives 0.40/0.337^2 ≈ 3.5 times (≈ 349 K) and STAC-3 b receives 0.08/0.182^2 ≈ 2.4 times (≈ 318 K). For comparison, Venus receives about 1.9 times Earth's flux.

We chose **STAC-3 b**. It is the only planet small enough to be rocky; planets larger than about 1.7 $R_\oplus$ usually keep thick hydrogen envelopes [6], which rules out STAC-1 b and STAC-2 b as worlds with a habitable surface. It also receives the least energy of the three and lies closest to its habitable zone, at 64% of d_{HZ}, compared with 53% for STAC-2 b and 5% for STAC-1 b. To sit inside the zone it would need a = 0.283 AU, which by (3) means a period of about 78 d rather than 40 d.

This choice has some real problems. A planet this close to an M dwarf is probably tidally locked and exposed to strong flares [7], and with T_{eq} near 318 K, surface water could only survive with help from clouds. The 170-day dip also still needs explaining.

Recommended observation. We would observe STAC-3 b with transmission spectroscopy on the James Webb Space Telescope [8]. During a transit some starlight passes through the planet's atmosphere, and gases such as H_2O, CO_2 and CH_4 absorb at particular wavelengths, making the transit slightly deeper at those wavelengths. We would combine this with radial-velocity measurements to get the planet's mass and density (a rocky planet should be around 5 g cm^{-3}), and with continuous photometry to pin down the period and check the 170-day event.

VI. Conclusion

All three stars show regular, equal-depth dips and so host transiting planets: a hot Neptune-size planet around STAC-1, a sub-Neptune around STAC-2 and a super-Earth around STAC-3. None of them lies in its habitable zone, but STAC-3 b is the smallest, the most likely to be rocky and the least strongly heated, so it is the planet we would study next.

References

[1] W. J. Borucki *et al.*, "Kepler planet-detection mission: Introduction and first results," *Science*, vol. 327, no. 5968, pp. 977–980, 2010.

[2] G. R. Ricker *et al.*, "Transiting Exoplanet Survey Satellite (TESS)," *J. Astron. Telesc. Instrum. Syst.*, vol. 1, no. 1, Art. 014003, 2015.

[3] S. Seager and G. Mallén-Ornelas, "A unique solution of planet and star parameters from an extrasolar planet transit light curve," *Astrophys. J.*, vol. 585, no. 2, pp. 1038–1055, 2003.

[4] J. N. Winn, "Transits and occultations," in *Exoplanets*, S. Seager, Ed. Tucson, AZ, USA: Univ. Arizona Press, 2010, pp. 55–77.

[5] J. F. Kasting, D. P. Whitmire, and R. T. Reynolds, "Habitable zones around main sequence stars," *Icarus*, vol. 101, no. 1, pp. 108–128, 1993.

[6] B. J. Fulton *et al.*, "The California-Kepler Survey. III. A gap in the radius distribution of small planets," *Astron. J.*, vol. 154, no. 3, Art. 109, 2017.

[7] A. L. Shields, S. Ballard, and J. A. Johnson, "The habitability of planets orbiting M-dwarf stars," *Phys. Rep.*, vol. 663, pp. 1–38, 2016.

[8] J. P. Gardner *et al.*, "The James Webb Space Telescope," *Space Sci. Rev.*, vol. 123, no. 4, pp. 485–606, 2006.